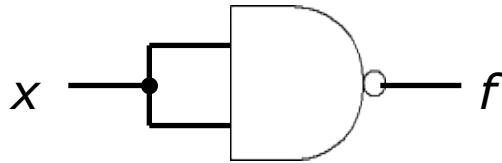


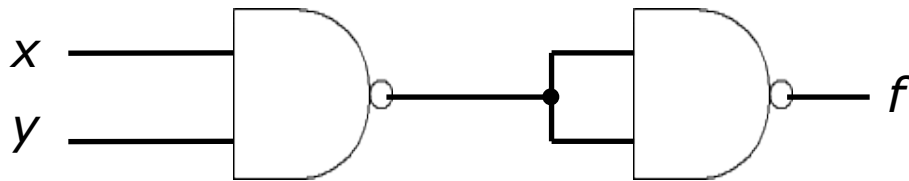
Gate Level Minimization

NAND Implementations of AND, OR, and NOT Gates



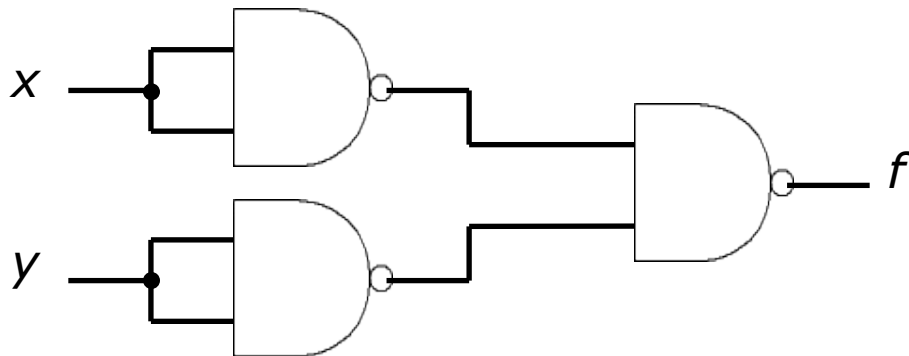
NOT Gate

$$f = x'$$



AND Gate

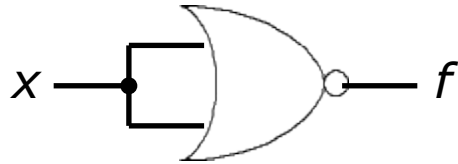
$$f = ((xy)')' = xy$$



OR Gate

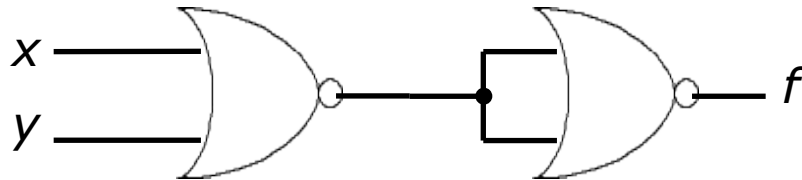
$$f = (x'y')' = x + y$$

NOR Implementations of AND, OR, and NOT Gates



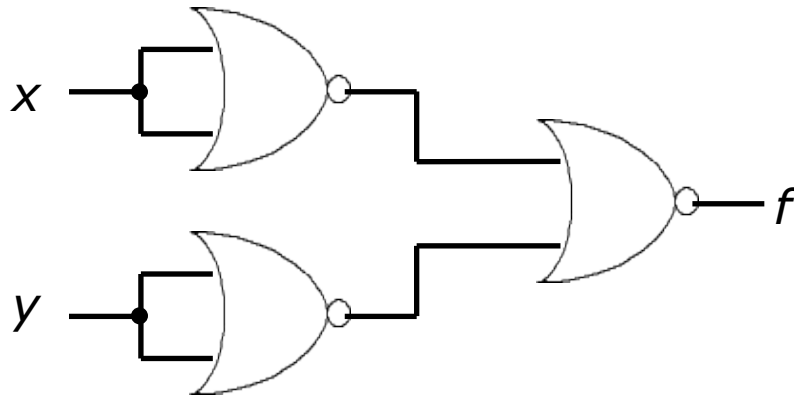
NOT Gate

$$f = x'$$



OR Gate

$$f = ((x + y)')' = x + y$$



AND Gate

$$f = (x' + y')' = xy$$

K-Map

- A Karnaugh map (K-map) is a pictorial method used to minimize Boolean expressions without having to use Boolean algebra.
- A K-map can be thought of as a special version of a truth table .
- It is used to simplify Boolean expressions where you have more than 2 variables

K-Map

2 Variable

		y	
x		0	1
	0		
	1		

K-Map

2 Variable

		y	
x		0	1
	0	00	01
	1	10	11

K-Map

2 Variable

		y	
x		0	1
	0	00 m_0	01 m_1
	1	10 m_2	11 m_3

Mapping in K-Map

- Each cell in a K-Map represents a minterm
- In the K-Map, put 1 in the cell where the corresponding minterm is 1; and zero otherwise.

Grouping and Minimizing

- Number of elements in a Groups must be 2^n
where n =whole number
- All 1s must be part of at least one group
- Group size must be maximized: e.g. if a group can be formed of four, do NOT make two groups (two in each).
- 1s can be shared among different groups
- Minimizing
 - For each group, select only those variables that are NOT changing their value throughout the group
 - Unchanged variables be named as x' or x representing 0 or 1 in that group

K-Map

3 Variable

		yz			
x		00	01	11	10
	0	000	001	011	010
	1	100	101	111	110

K-Map

3 Variable

		yz			
x		00	01	11	10
	0	000 m_0	001 m_1	011 m_3	010 m_2
	1	100 m_4	101 m_5	111 m_7	110 m_6

K-Map

3 Variable

- $F(x,y,z) = \Sigma(2,3,4,5)$

$$F = x'y + xy'$$

		yz				
			00	01	11	10
x	0				1	1
	1	1	1			

K-Map

3 Variable

- $F(x,y,z) = \Sigma(3,4,6,7)$

$$F = yz + xz'$$

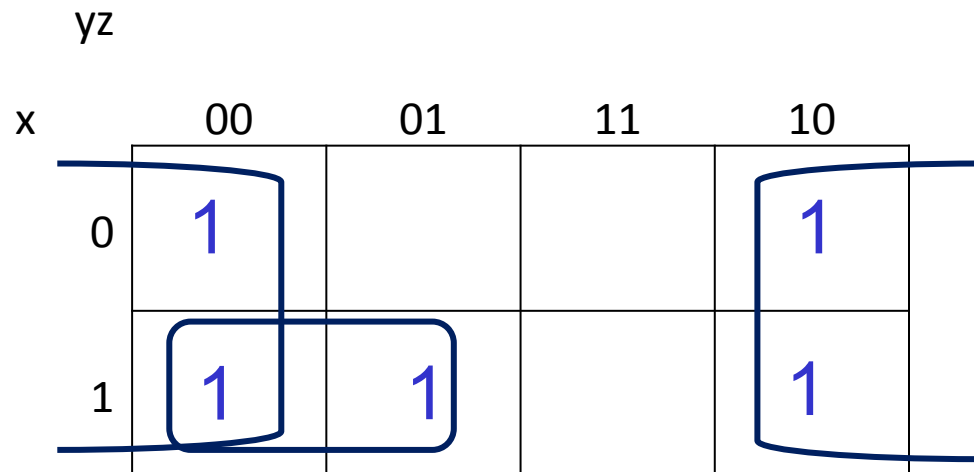
		yz			
		00	01	11	10
x	0			1	
	1	1		1	1

K-Map

3 Variable

- $F(x,y,z) = \Sigma(0,2,4,5,6)$

$$F = z' + xy'$$



K-Map

3 Variable

- $F(A,B,C) = A'C + A'B + AB'C + BC$

$$F = C + A'B$$

BC

A	00	01	11	10
0		1	1	1
1		1	1	

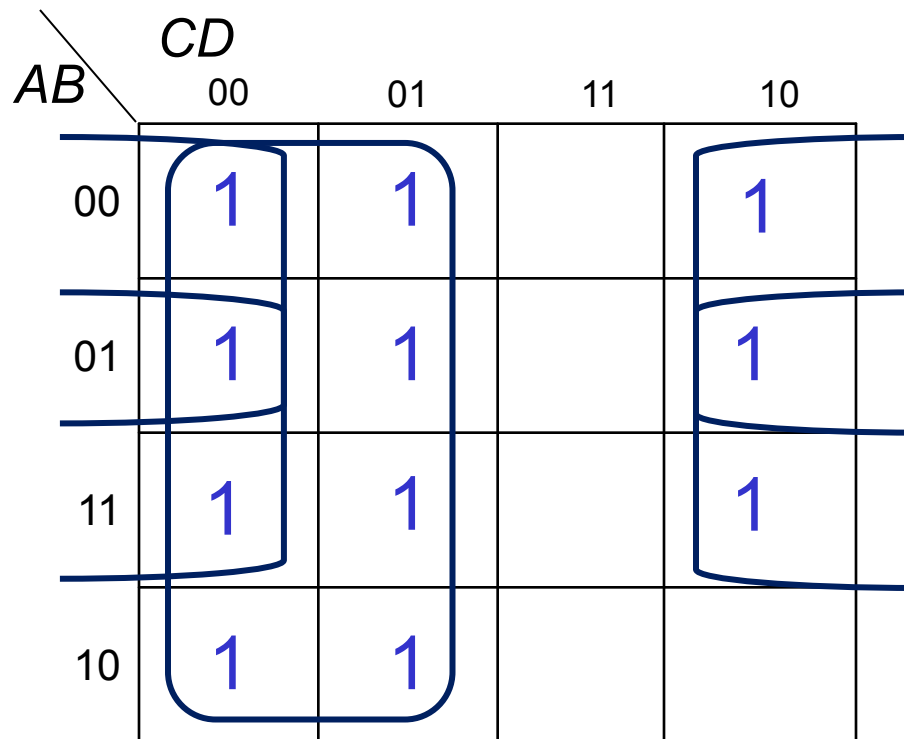
Four Variable Maps

m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6
m_{12}	m_{13}	m_{15}	m_{14}
m_8	m_9	m_{11}	m_{10}

		y			
		00	01	11	10
w	yz				
	00	m_0 $w'x'y'z'$	m_1 $w'x'y'z$	m_3 $w'x'yz$	m_2 $w'x'yz'$
	01	m_4 $w'xy'z'$	m_5 $w'xy'z$	m_7 $w'xyz$	m_6 $w'xyz'$
	11	m_{12} $wxy'z'$	m_{13} $wxy'z$	m_{15} $wxyz$	m_{14} $wxyz'$
	10	m_8 $wx'y'z'$	m_9 $wx'y'z$	m_{11} $wx'yz$	m_{10} $wx'yz'$
		z			

Four Variable Maps

- $F(A,B,C,D) = \Sigma(0,1,2,4,5,6,8,9,12,13,14)$



$$F = C' + A'D' + BD'$$

Four Variable Maps

- $F(A,B,C,D) = \Sigma(0,1,4,5,10,11)$

<i>AB</i> \ <i>CD</i>	<i>CD</i>			
	00	01	11	10
00				
01				
11				
10				

Four Variable Maps

- $F(A,B,C,D) = \Sigma(0,2,3,5,7,8,9,10,11,13,15)$

<i>AB</i> \ <i>CD</i>	<i>CD</i>			
	00	01	11	10
00				
01				
11				
10				

Prime Implicants & Essential Prime Implicants

- A prime implicant is product term obtained by combining the maximum possible number of adjacent squares in the map.
- If a minterm in a square/cell is covered by only one prime implicant, that prime implicant is said to be essential.

Prime Implicants & Essential Prime Implicants - Example

$$F(A,B,C) = \Sigma(1,5)$$

Prime Implicant = $B'C$

BC					
A		00	01	11	10
0			1		
1			1		

Prime Implicants & Essential Prime Implicants - Example

$$F(A,B,C) = \Sigma(1,3,5,7)$$

~~B'C is NOT a Prime Implicant~~

Prime Implicant = C

		BC			
A		00	01	11	10
	0		1	1	
	1		1	1	

Prime Implicants & Essential Prime Implicants - Example

$$F(A,B,C) = \Sigma(1,3,4,5)$$

Prime Implicants

$B'C$

$A'C$

AB'

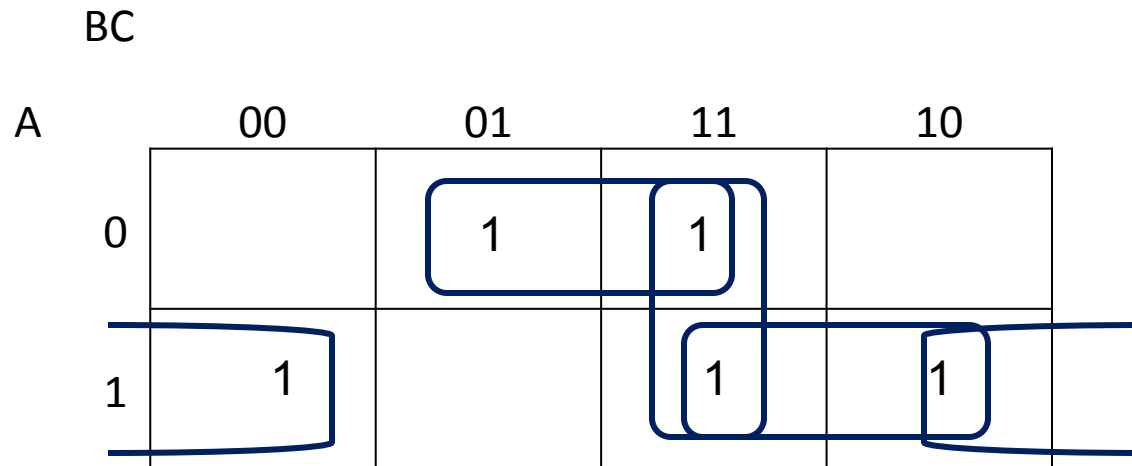
BC					
A		00	01	11	10
0			1	1	
1		1	1		

Essential Prime Implicant = $A'C$ & AB'

Prime Implicants & Essential Prime Implicants - Example

Find all the Prime Implicants in the following equation and identify which are essential

$$F(A,B,C) = \Sigma(1,3,4,6,7)$$



Prime Implicants

A'C

AC'

AB

BC

Essential Prime Implicant = A'C & AC'

Example

- $F(A,B,C,D) = \Sigma(0,2,3,5,7,8,9,10,11,13,15)$

<i>AB</i> \ <i>CD</i>	<i>CD</i>			
	00	01	11	10
00	1		1	1
01		1	1	
11		1	1	
10	1	1	1	1

Example

- $F(A,B,C,D) = \Sigma(0,2,3,5,7,8,9,10,11,13,15)$

<i>AB</i> \ <i>CD</i>	<i>CD</i>			
	00	01	11	10
00	1		1	1
01		1	1	
11		1	1	
10	1	1	1	1

Example

- $F(A,B,C,D) = \Sigma(0,2,3,5,7,8,9,10,11,13,15)$

<i>AB</i> \ <i>CD</i>	<i>CD</i>			
	00	01	11	10
00	1		1	1
01		1	1	
11		1	1	
10	1	1	1	1

Example

- $F(A,B,C,D) = \Sigma(0,2,3,5,7,8,9,10,11,13,15)$

<i>AB</i> \ <i>CD</i>	<i>CD</i>			
	00	01	11	10
00	1		1	1
01		1	1	
11		1	1	
10	1	1	1	1

Example

- $F(A,B,C,D) = \Sigma(0,2,3,5,7,8,9,10,11,13,15)$

<i>AB</i> \ <i>CD</i>	<i>CD</i>			
	00	01	11	10
00	1		1	1
01		1	1	
11		1	1	
10	1	1	1	1

Example

- $F(A,B,C,D) = \Sigma(0,2,3,5,7,8,9,10,11,13,15)$

$AB \backslash CD$		CD	
		00	01
00	1		1
01		1	1
11		1	1
10	1	1	1

Example

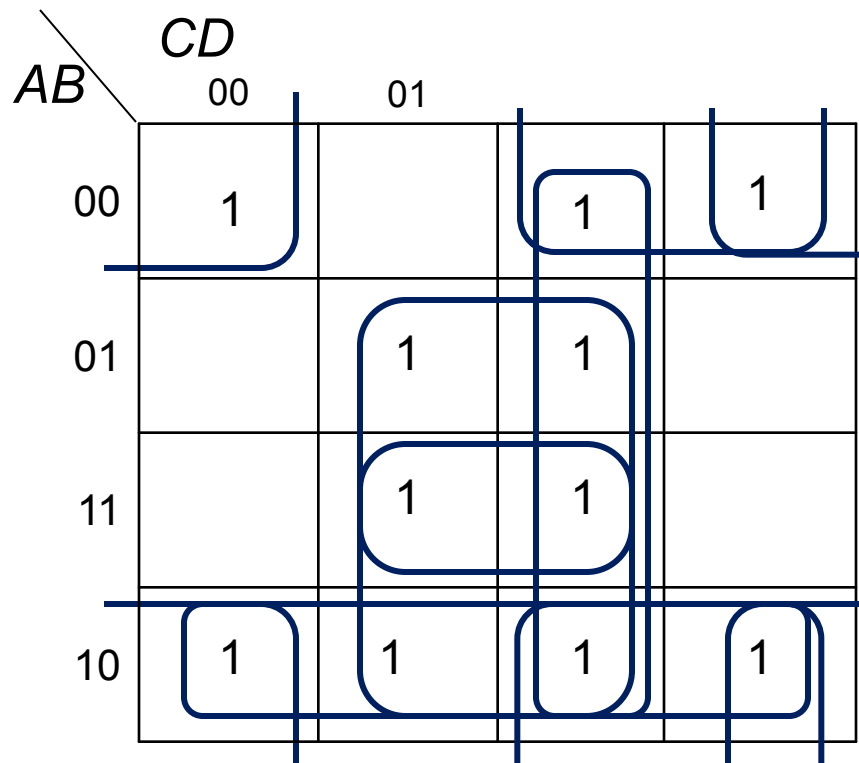
- $F(A,B,C,D) = \Sigma(0,2,3,5,7,8,9,10,11,13,15)$

A 4x4 Karnaugh map for the function F(A,B,C,D). The vertical axis is labeled AB with values 00, 01, 11, 10. The horizontal axis is labeled CD with values 00, 01. The map contains 1s in the following cells: (00,00), (00,10), (01,01), (01,11), (10,00), (10,01), (10,11), (10,10), (11,01), (11,11), (11,10), (11,00). There are four groups of 1s circled with blue lines: a group of four 1s in the first row (AB=00), a group of four 1s in the first column (CD=00), a group of four 1s in the last column (CD=10), and a group of four 1s in the last row (AB=10).

$AB \backslash CD$	00	01	11	10
00	1		1	1
01		1	1	
11		1	1	
10	1	1	1	1

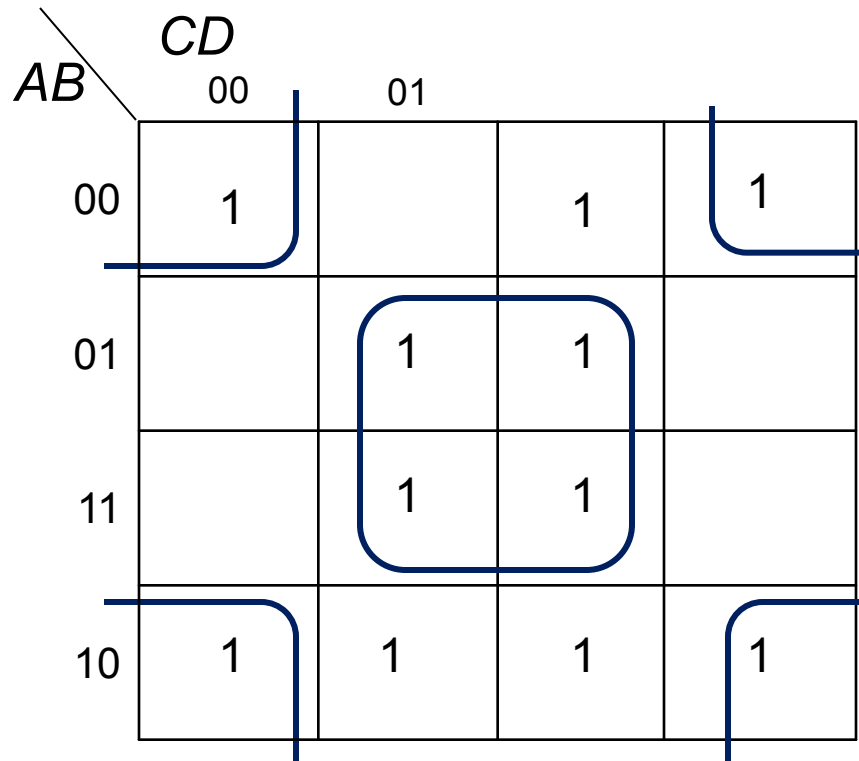
Example

- $F(A,B,C,D) = \Sigma(0,2,3,5,7,8,9,10,11,13,15)$



Example

- $F(A,B,C,D) = \Sigma(0,2,3,5,7,8,9,10,11,13,15)$



		CD			
		00			
AB		01	11	10	
	00	1			1
	01		1	1	
	11		1	1	
	10	1			1

(a) Essential Prime Implicants BD and $B'D'$

		CD			
		00			
AB		01	11	10	
	00	1		1	1
	01		1	1	
	11		1	1	
	10	1		1	1

(b) Prime implicants, CD , $B'C$, AD , and AB'

Fig 3-11

Four Possible solutions

- $F = BD + B'D' + CD + AD$
 - $F = BD + B'D' + CD + AB'$
 - $F = BD + B'D' + B'C + AD$
 - $F = BD + B'D' + B'C + AB'$
-
- *All solutions include the EPIs.*

Example

- $F(A,B,C,D) = \Sigma(0,2,4,5,6,7,8,10,13,15)$

<i>AB</i> \ <i>CD</i>	<i>CD</i>			
	00	01	11	10
00				
01				
11				
10				

Five Variable Map

- Five-Variable map consists of 2 four-variable maps with variable A, B, C, D and E.
- Variable A distinguished between the two maps, as indicated on the top of the diagram.

Five Variable Map

$A = 0$

$BC \backslash DE$	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$A = 1$

$BC \backslash DE$	00	01	11	10
00	16	17	19	18
01	20	21	23	22
11	28	29	31	30
10	24	25	27	26

Fig 3-12

Example:

$$F(A,B,C,D,E) = \Sigma(0,2,4,6,9,13,21,23,25,29,31)$$

$A = 0$

$BC \backslash DE$	00	01	11	10
00				
01				
11				
10				

$A = 1$

$BC \backslash DE$	00	01	11	10
00				
01				
11				
10				

Fig 3-12

Example:

$$F(A,B,C,D,E) = \Sigma(0,2,4,6,9,13,21,23,25,29,31)$$

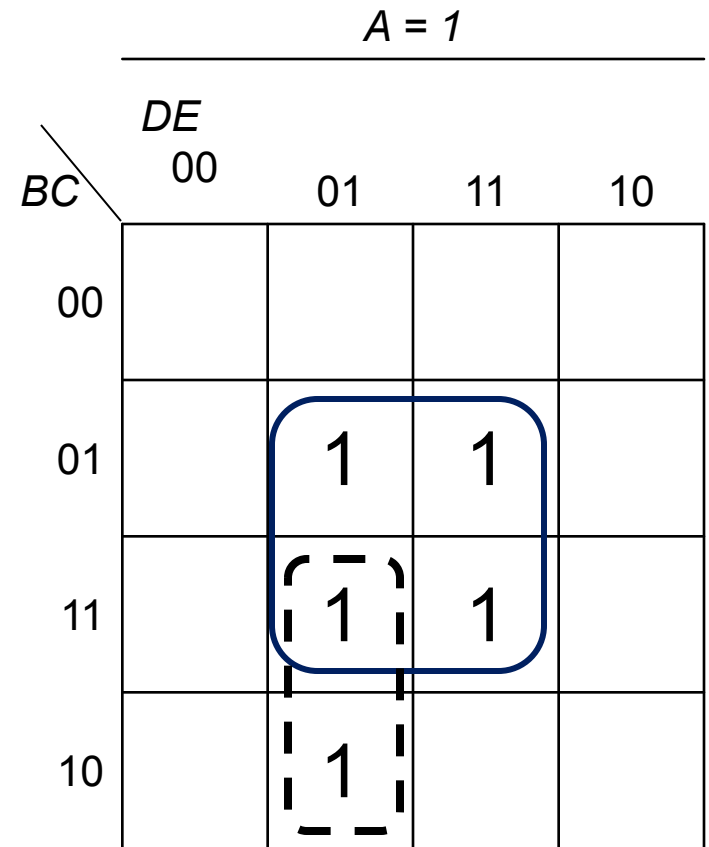
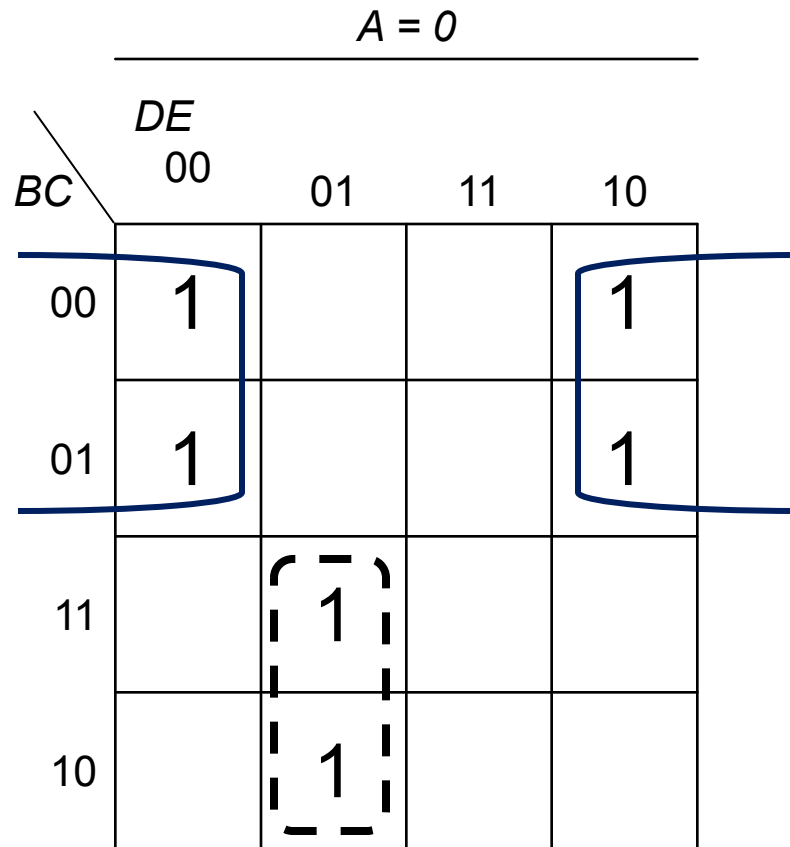


Fig 3-12

$$F = A'B'E' + BD'E + ACE$$

Example:

$$F(A,B,C,D,E) = \Sigma(0,1,4,5,16,17,21,25,29)$$

3-11a

$A = 0$

$BC \backslash DE$	00	01	11	10
00				
01				
11				
10				

$A = 1$

$BC \backslash DE$	00	01	11	10
00				
01				
11				
10				

$$F = A'B'D' + AD'E + B'C'D'$$

$A = 0$				
$BC \backslash DE$	00	01	11	10
00				
01				
11				
10				

$A = 1$				
$BC \backslash DE$	00	01	11	10
00				
01				
11				
10				

Product of Sums

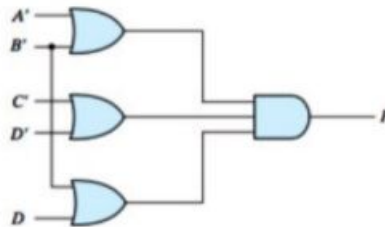
Simplify the Boolean function $F(A,B,C,D) = \sum (0,1,2,5,8,9,10)$ into **product-of-sums** form.

For a function F , combining 0 squares gives us F' .

$$\overline{F} = \overline{B}\overline{D} + CD + AB$$

By using DeMorgan's Law, we can simplify the function to product-of-sums.

$$F = (\overline{B} + D)(\overline{C} + \overline{D})(\overline{A} + \overline{B})$$



Truth table for $F(A,B,C,D)$ showing the function value for all combinations of A, B, C, D . The table is a 4x4 grid with AB as rows and CD as columns. Red boxes highlight the 0s in the table, which correspond to the minterms in the sum of products for F' .

		CD			
		00	01	11	10
AB	00	1	1	0	1
	01	0	1	0	0
	11	0	0	0	0
	10	1	1	0	1

Don't Care Conditions

Don't Care Conditions

- $F(w,x,y,z) = \sum (1,3,7,11,15) + d(w,x,y,z) = \sum (0,2,5)$

$$F = yz + w'z$$

WX \ yz	00	01	11	10
00	X	1	1	X
01		X	1	
11			1	
10			1	

Don't Care Conditions

- $F(w,x,y,z) = \sum (1,3,7,11,15) +$
 $d(w,x,y,z) = \sum (0,2,5)$

$$F = yz + w'x'$$

$wx \backslash yz$	00	01	11	10
00	X	1	1	X
01		X	1	
11			1	
10			1	

Don't Care Conditions (Example)

- $F(w,x,y,z) = \sum (0,1,2,4,5) + d(w,x,y,z) = \sum (3,6,7)$

WX \ yz	yz			
	00	01	11	10
00	1	1	X	1
01	1	1	X	X
11				
10				

$$F = w'$$

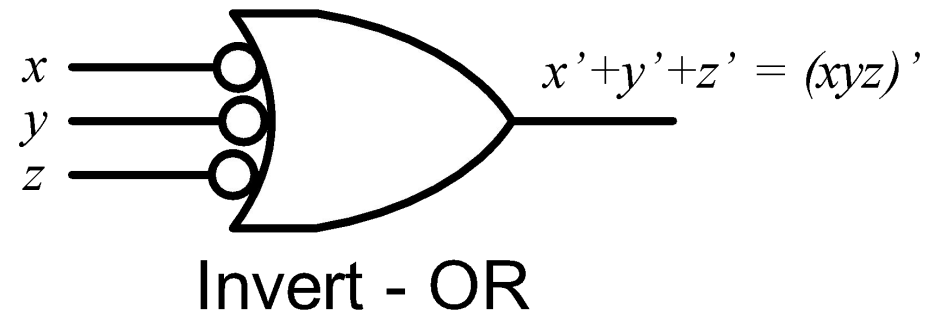
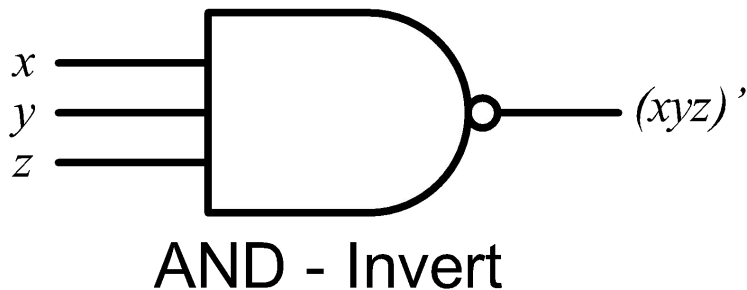
NAND implementation

NAND Implementation

- Digital circuits are frequently constructed with NAND or NOR gates
- These gates are easier to fabricate
- NAND as universal gate (practice in lab)

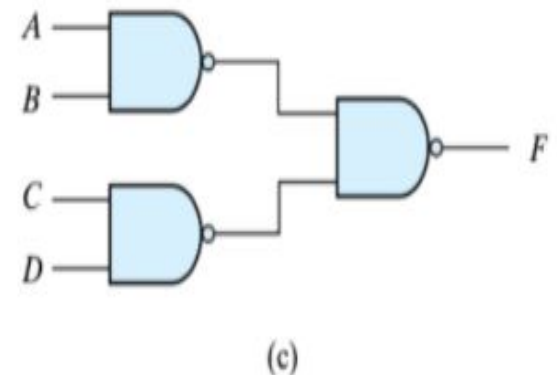
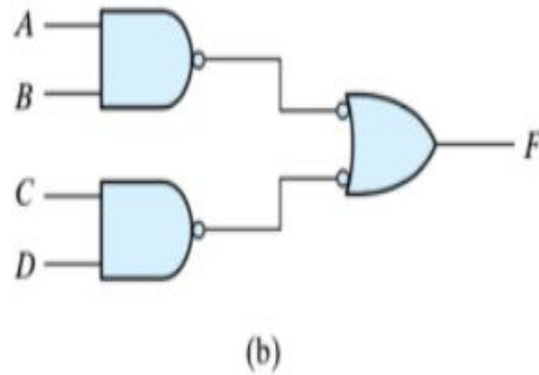
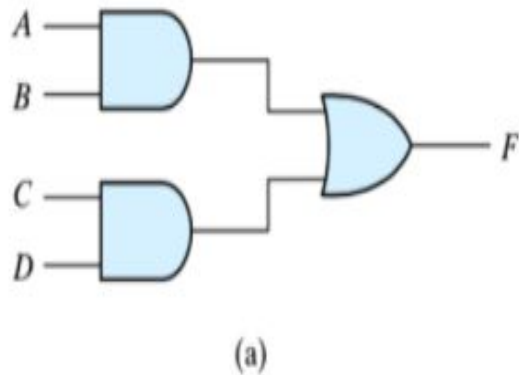
NAND Implementation

- Two Graphic Symbols



NAND-Two Level Implementation

- Three ways to implement $(A.B) + (C.D)$



NAND – Two Level Implementation

1. (Simplified) Equation must be in Sum-of-Products form. (Two Levels)
2. Draw AND-OR diagram
Assume all variables and their complements are available. i.e. do not use inverters to take complement of a variable.
3. Replace AND gates with AND-Invert symbol.
Replace OR gates with Invert-OR symbols

NAND – Two Level Implementation

- Example

- $F(x,y,z) = \Sigma(1,2,3,4,5,7)$

- 1. Step-1: Simplify and express it in Sum of Products form

- Use K-Map to simplify and convert into sum-of-product form

- $F = xy' + x'y + z$

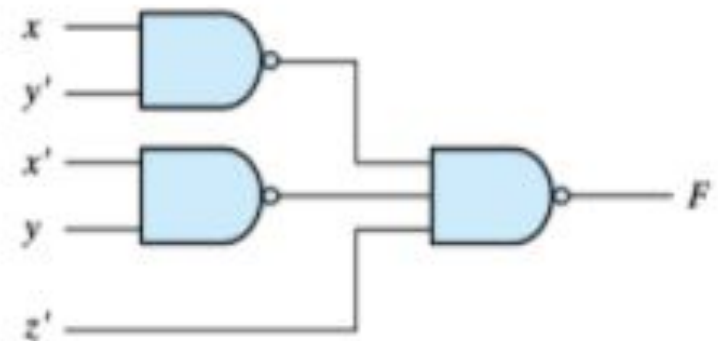
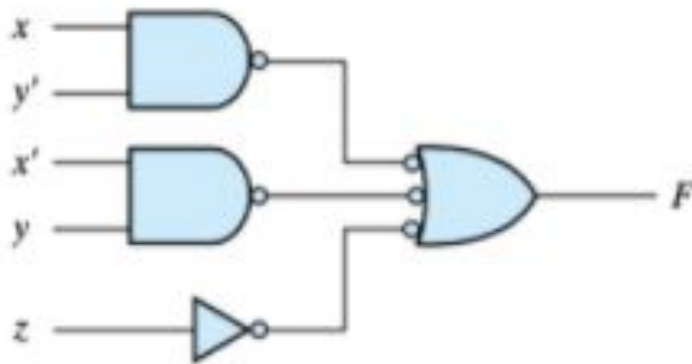
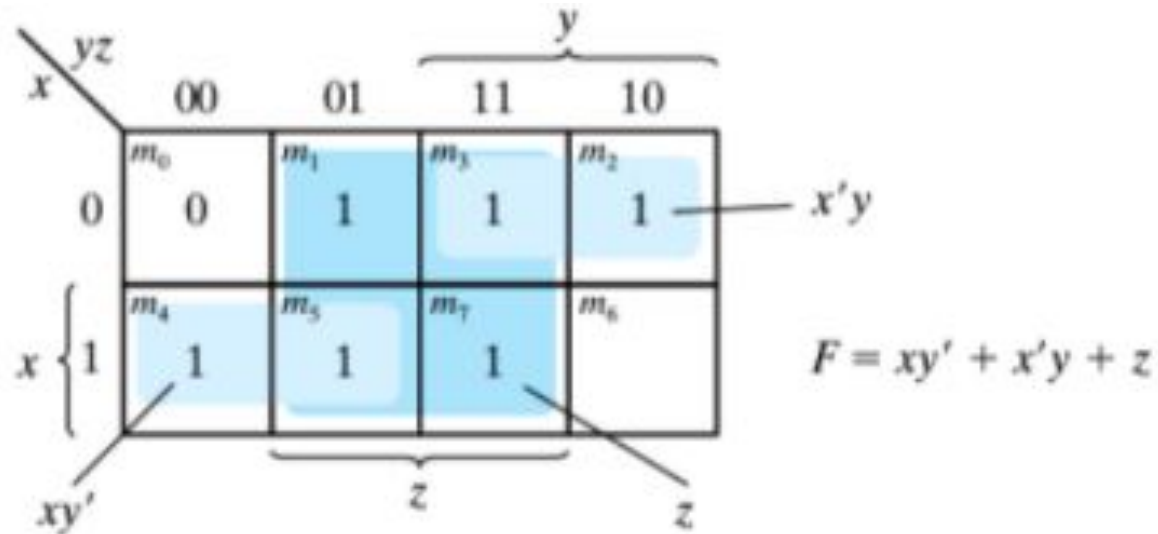
- 2. *Draw an AND-OR diagram*

- 3. *Replace AND gates with AND-Invert symbols*

- *Replace OR gates with Invert-OR symbols*

- Invert the symbol going into the second level*

NAND-Two Level Implementation



Multilevel NAND Circuit

- If three or more levels in the equation, follow these steps.
 1. Draw a simple AND-OR diagram
Assume all variables and their complements are available. i.e. do not use inverters to take complement of a variable.
 2. Convert all AND gates to NAND gates with AND-Invert Graphic symbols
Convert all OR gates to NAND gates with Invert-OR Graphic symbols
 3. Check for all the bubbles in the diagram (i.e. must have two bubbles per connecting line). Compensate all bubbles (with inverter)

Multilevel NAND Circuit

- Example:

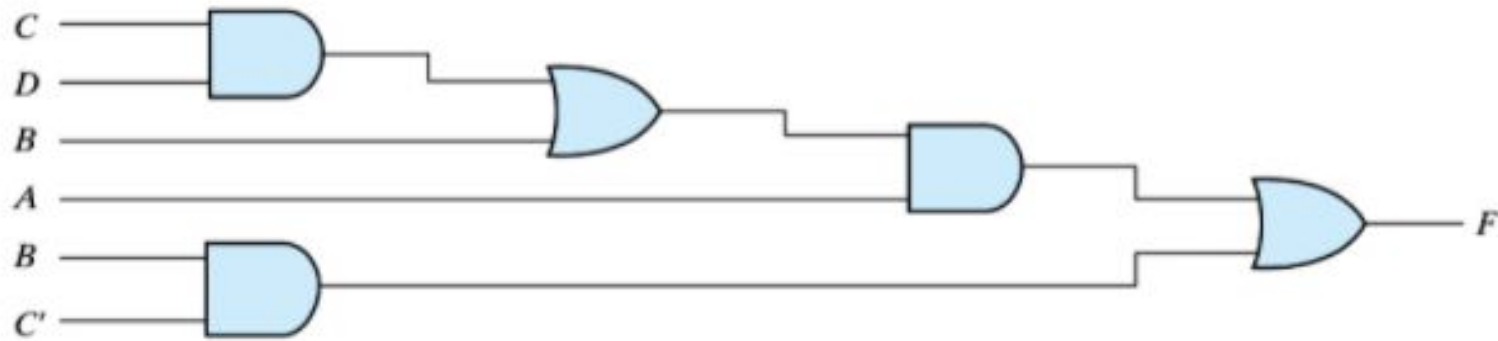
- $$F = A(CD + B) + BC'$$

- Step-1: Draw AND-OR Diagram

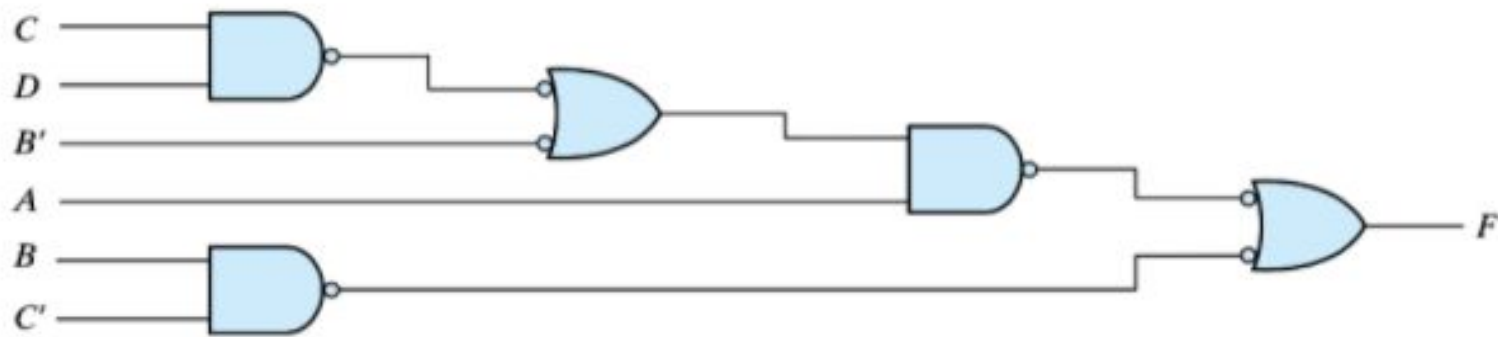
- Step-2: AND to NAND, OR to Invert-OR

- Step-3: Compensate Bubbles

Multilevel NAND Circuit



(a) AND-OR gates



(b) NAND gates

Multilevel NAND Circuit

- Examples
- $F = (AB' + A'B)(C + D')$

Multilevel NAND Circuit

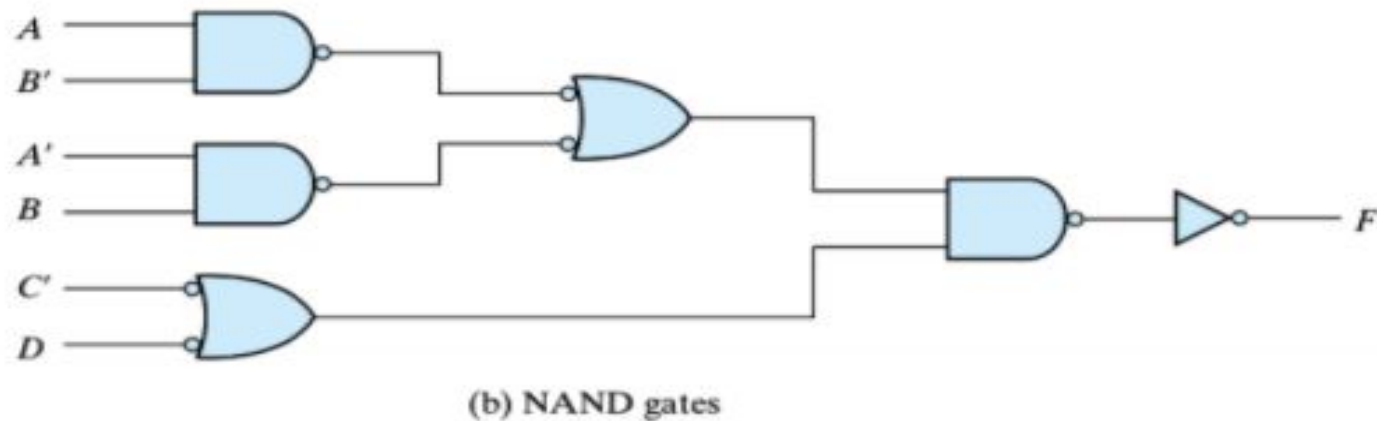
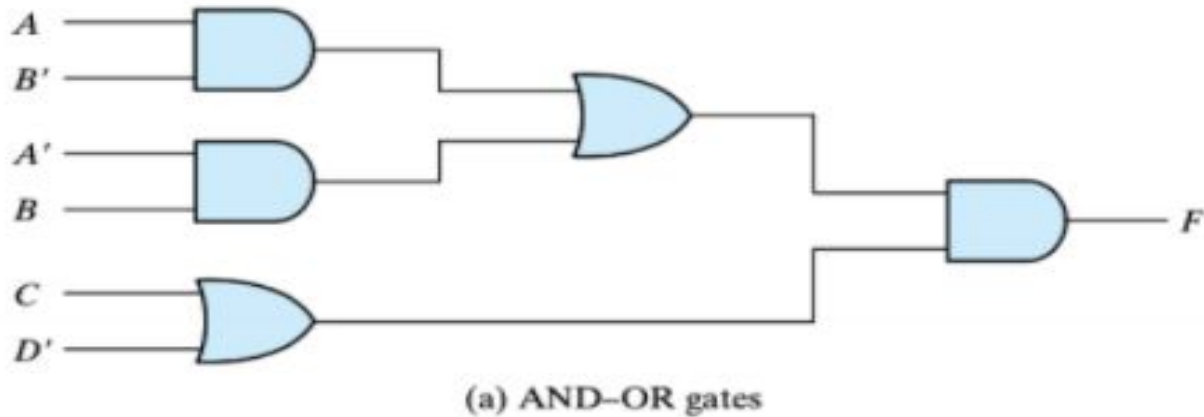


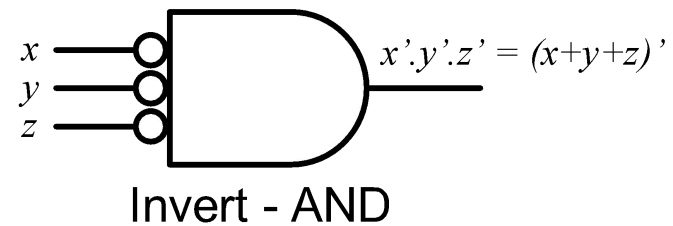
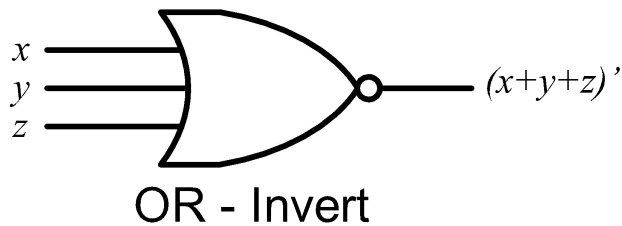
FIGURE 3.21

Implementing $F = (AB' + A'B)(C + D')$

NOR Implementation

NOR Implementation

- Two Graphic Symbols



NOR – Two Level Implementation

1. Equation must be in Product-of-Sum form. (Two Levels)
2. Draw a AND-OR diagram
3. Replace OR gates with OR-Invert symbol.
Replace AND gates with Invert-AND symbols
4. If an input is directly going into the second level, invert it

NOR – Two Level Implementation

- Example
- $F = (A + B)(C + D)E$

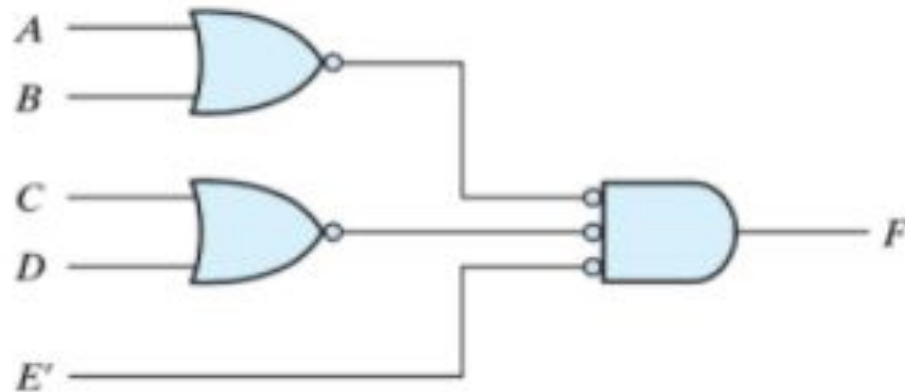


FIGURE 3.24

Implementing $F = (A + B)(C + D)E$

Multilevel NOR Circuit

- If three or more levels in the equation, follow these steps.
 1. Draw a simple AND-OR diagram
 2. Convert all OR gates to NOR gates with
OR-Invert Graphic symbols
Convert all AND gates to NAND gates with Invert-AND Graphic symbols
 3. Check for all the bubbles in the diagram. Compensate all bubbles (with inverter)

Multilevel NOR Circuit

- Example:
- $F = (AB' + A'B)(C + D')$

Selected Problems

- Simplify the following expression and implement with two-level NAND circuit

- $F = AB' + ABD + ABD' + A'C'D' + A'BC'$

- Step-1: Simplified circuit be in SOP $F = A + BC' + C'D'$ form

AB \ CD	CD			
	00	01	11	10
00	1			
01	1	1		
11	1	1	1	1
10	1	1	1	1

- Simplify the following expression and implement with two-level NAND circuit
- Step-1: *Simplified circuit be in SOP form*
 - $F = A + BC' + C'D'$
- Step-2: *Draw AND-OR Diagram*
- Step-3: *Replace AND with AND-Invert
OR with Invert-OR*
- Step-4: *Input A is going directly to second level. Invert it.*

- Problem 3-19: Simplify the following expression and implement with two-level NOR circuit
- $F(w, x, y, z) = \Sigma(5, 6, 9, 10)$
- Step-1: Simplified circuit be in POS form

WX

	00	01	11	10
00	0	0	0	0
01	0	1	0	1
11	0	0	0	0
10	0	1	0	1

$$F' = w'x' + wx + y'z' + yz$$

$$F = (w+x)(w'+x')(y+z)(y'+z')$$

- Problem 3-20: Draw the multiple-level NAND circuit for the following expression
- $(AB' + CD')E + BC(A+B)$
- Step-1: Draw AND-OR Diagram
- Step-2: AND to NAND, OR to Invert-OR
- Step-3: Compensate Bubbles

Problem 3-22: Convert the following logic diagram into multiple-level NAND circuit

